

MILK RIVER, AB, Canada

WMO: 712440

Lat: 49.134N

Lon: 112.051W

Elev: 3445

StdP: 12.96

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-13.3	-26.5	1.4	-19.9	-19.2	2.2	-13.2	36.2	38.9	32.3	35.2	7.5	300	0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.9	89.3	60.2	85.9	59.5	82.4	59.0	64.0	79.9	62.3	78.8	60.8	77.2	10.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	84.9	68.1	56.8	78.1	66.6	54.7	72.4	65.0	31.0	79.6	29.7	78.7	28.5	77.1	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.8	26.0	22.8	-29.4	95.9	6.6	3.3	-34.2	98.3	-38.0	100.2	-41.7	102.1	-46.5	104.5
			-29.6	69.0	6.6	5.6	-34.3	73.0	-38.2	76.3	-42.0	79.4	-46.8	83.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	20.0	22.2	30.2	40.5	49.9	57.4	64.9	63.5	54.2	41.8	29.6	20.6	
	DBStd	19.41	15.97	14.51	13.68	8.93	7.88	6.16	5.82	6.23	8.28	9.67	13.55	14.74	
	HDD50	4593	931	779	614	300	99	10	1	2	49	281	615	912	
	HDD65	8784	1396	1199	1078	735	470	240	74	103	333	718	1061	1377	
	CDD50	1431	0	0	1	15	96	231	461	420	176	27	4	0	
	CDD65	148	0	0	0	0	2	10	70	56	10	0	0	0	
	CDH74	3534	0	0	0	7	113	278	1434	1300	386	16	0	0	
	CDH80	1246	0	0	0	0	17	62	548	496	121	2	0	0	
Wind	WSAvg	10.4	11.0	10.5	11.2	11.5	10.9	10.0	8.9	8.6	9.4	10.7	10.8	10.8	
Precipitation	PrecAvg	13.3	0.4	0.3	0.6	1.3	1.9	3.3	1.1	1.3	1.4	0.7	0.4	0.4	
	PrecMax	20.2	0.7	1.0	1.7	2.4	3.9	10.7	2.4	3.4	3.9	1.9	1.1	0.8	
	PrecMin	7.4	0.1	0.1	0.2	0.4	0.4	0.5	0.1	0.2	0.2	0.1	0.1	0.1	
	PrecStd	3.2	0.2	0.2	0.4	0.7	1.0	2.3	0.8	0.8	0.9	0.5	0.2	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	56.1	63.8	74.4	82.2	87.0	94.8	93.2	88.9	76.0	64.8	53.2	
		MCWB	41.7	41.8	46.0	51.2	57.2	62.6	61.4	60.2	57.9	53.6	47.9	40.9	
	2%	DB	46.7	49.1	58.5	68.3	77.2	80.9	89.8	89.3	83.6	69.2	57.6	46.8	
		MCWB	38.8	38.7	43.7	48.8	54.9	59.1	61.1	59.3	56.4	50.8	44.6	38.2	
	5%	DB	43.2	44.7	53.7	62.6	72.1	76.9	86.1	85.7	78.5	63.5	52.4	43.1	
		MCWB	36.5	36.6	41.6	45.9	52.7	57.8	60.8	58.9	55.2	48.1	41.5	36.0	
	10%	DB	40.3	40.9	48.8	57.5	67.2	72.9	82.4	81.9	72.7	58.2	47.5	39.8	
		MCWB	34.8	34.0	38.8	43.3	50.7	56.2	60.3	58.3	53.2	45.3	38.8	33.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.7	43.3	47.6	52.5	58.8	65.2	67.9	64.4	60.9	55.3	49.4	42.3
			MCDB	50.3	54.4	61.2	69.8	76.7	79.5	83.3	80.6	78.7	72.3	61.7	50.6
2%		WB	39.2	39.6	44.6	49.5	56.2	62.0	65.0	62.6	58.2	51.6	45.1	38.9	
		MCDB	45.8	48.1	56.4	66.6	73.0	75.9	80.4	80.4	77.2	66.3	56.5	46.1	
5%		WB	37.0	36.9	42.1	46.8	54.1	59.8	63.2	61.1	56.2	48.8	42.2	36.4	
		MCDB	42.6	44.2	52.6	61.0	69.1	72.8	79.4	78.9	74.9	61.9	51.4	42.6	
10%		WB	34.9	34.5	39.3	44.1	51.9	57.7	61.6	59.6	54.2	46.2	39.3	34.0	
		MCDB	40.1	40.5	48.2	55.9	65.3	69.9	78.0	77.2	71.4	57.3	46.8	39.6	
Mean Daily Temperature Range	5% DB	MDBR	21.0	21.6	22.3	24.1	25.0	24.8	29.9	30.6	28.2	25.3	21.5	21.0	
		MCDBR	21.5	25.2	29.4	34.0	34.1	32.4	36.8	38.8	38.7	33.3	26.8	22.7	
		MCWBR	15.0	16.2	16.9	17.2	15.5	14.6	13.9	14.4	16.2	17.4	15.5	15.5	
	5% WB	MCDBR	21.3	24.4	28.2	31.6	30.2	28.5	31.3	32.4	34.3	30.8	25.6	22.2	
		MCWBR	15.6	16.4	16.9	16.6	14.5	14.3	13.9	13.7	15.5	16.7	15.5	15.7	
Clear-Sky Solar Irradiance	taub	0.252	0.260	0.279	0.317	0.331	0.331	0.333	0.351	0.313	0.282	0.255	0.242		
	taud	2.468	2.494	2.498	2.396	2.402	2.453	2.456	2.378	2.456	2.529	2.511	2.453		
	Ebn at Noon	266	290	300	296	294	294	292	281	283	277	263	254		
	Edh at Noon	19	23	28	34	36	34	34	35	29	22	18	16		
All-Sky Solar Radiation	RadAvg	400	713	1110	1536	1828	2044	2256	1812	1290	801	449	317		
	RadStd	33	56	93	99	151	141	123	115	104	77	32	29		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page